

Basics of Communication Systems—Homework 2

Due date: 3/20

Let $m(t) = 0.1 + 3 \cos\left(2\pi f_m \frac{(t-1.5)}{25}\right) - 4 \sin\left(2\pi f_m \frac{(t-0.5)}{15}\right)$ be a continuous-time signal, where $f_m = 100$.

1. Sample $m(t)$ with sampling frequency $f_s = 300$. Plot the sampled signal overlaid on $m(t)$.
2. Quantize the samples using a uniform quantizer with L quantization levels. Plot the quantized samples with $L = 4, 16, 64$. (the value of the smallest quantization level is $\min_{0 \leq t \leq 1} m(t)$ and the value of the largest quantization level is $\max_{0 \leq t \leq 1} m(t)$)
3. With $L = 16$, form a pulse-amplitude-modulated (PAM) signal using rectangular pulses of duration $T_s = 1/f_s$.
4. Add white Gaussian noise with variance 0.3 to the PAM signal. Plot the noisy received signal.
5. Pass the received signal through a matched filter and plot the output signal.
6. Sample the filter output at symbol instants and obtain detected symbol values $\hat{a}_n$'s. Plot the detected symbol values.
7. Reconstruct the analog signal from detected samples $\hat{a}_n$'s using square wave interpolation. Plot the reconstructed signal $\hat{m}(t)$ together with the original message $m(t)$.
8. (optional) Reconstruct the analog signal from detected samples using sinc interpolation

$$\hat{m}(t) = \sum_n \hat{a}_n \text{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Plot the reconstructed signal $\hat{m}(t)$ together with the original message $m(t)$.

9. Simulate an analog AM transmission of $m(t)$ using DSC-C with carrier frequency $f_c = 1000$ and modulation index $k_a = 0.7$ over the same noisy channel. Plot and compare the recovered signals from DSC-C and PAM transmission.
10. Explain why digital communication systems are more robust to noise.